BCAC 0025: COMPUTER SYSTEM ARCHITECTURE

Objective: This course is designed to provide an introduction to the Computer System Architecture.

Credits: 04 **L–T–P: 4–0–0**

Module No.	Content	Teaching Hours
I	**Data Representation and basic Computer Arithmetic:** Number systems, complements, fixed and floating point representation, character representation, addition, subtraction, magnitude comparison. **Logic gates and circuits:** logic gates, boolean algebra, combinational circuits, circuit simplification, introduction to flip-flops and sequential circuits, decoders, multiplexers, registers, counters. **Basic Computer Organization and Design:** Computer registers, bus system, instruction set, timing and control, instruction cycle, memory reference, input-output and interrupt. **Central Processing Unit:** Register organization, arithmetic and logical micro-operations, stack organization, Hardwired vs. micro programmed control. Pipeline control: Instruction pipelines, pipeline performance, super scalar processing, Pipelining, RISC & CISC.	30
II	**Programming the Basic Computer:** Instruction formats, addressing modes, instruction codes, assembly language. **Memory Organization:** Memory device characteristics, random access memories, serial access memories, Multilevel memories, address translation, memory allocation, Main features, address mapping, structure versus performance. **Input-output Organization:** Peripheral devices, I/O interface, Modes of data transfer: Programmed, Interrupt Driven and Direct Memory Access. **Parallel processing:** Processor-level parallelism, multiprocessor architecture.	30

Text Book:

- M. Mano, “Computer System Architecture”, Pearson Education, New Jersey, 2017,Third Edition.

Reference Books:

- W. Stallings, “Computer Organization and Architecture Designing for Performance”, Prentice Hall of India, 2015, Tenth Edition.
- M. Mano, “Digital Design”, Pearson Education, New Jersey, 2018, Sixth Edition.
- Vranasic and Hamacher, Computer Organization, TMH"

Focus: This course focuses on Employability under CO2, CO3, CO4 and CO5.

Outcome: The student will be able to understand

CO1: Understand the basic arithmetic of a Computer System; how the data is represented,

CO2: Evaluate the various operations are performed on the data, the basic circuits to perform these operations,

CO3: Analyze how instructions are formatted and how these instructions are executed to accomplish a particular operation.

CO4: Explore the organization of the peripheral devices, the interface between these devices to the system.

CO5: Understand the architecture of a basic computer, its registers, bus system and the interaction flow among them.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3/PSO1
CO2	PO1,PO3/PSO1
CO3	PO2,PO3,PO5/PSO2
CO4	PO2,PO3,PO4/PSO1,PSO3
CO5	PO2,PO3/PSO2